

EFY Note

The source code of this project is available for free download at source.efymag.com

ISS_LED_SAMPLE | Arduino 1.8.9

File Edit Sketch Tools Help

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ISS_LED_SAMPLE

#include <ESP8266WiFi.h>
#include "Adafruit_MQTT.h"
#include "Adafruit_MQTT_Client.h"

#define WIFI_SSID ".....SSID....."//Your wifi name
#define WIFI_PASS ".....PASSWORD....."//your wifi passw

#define MQTT_SERV "io.adafruit.com"
#define MQTT_PORT 1883
#define MQTT_NAME ".....Adafruit ID....." //Your adafru:
#define MQTT_PASS ".....AIO KEY....." //Your adaf:

int led = D7;

```

Fig. 11: The sample code that needs to be modified

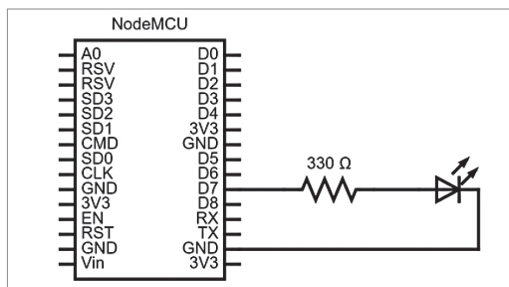


Fig. 12: NodeMCU connections

USERNAME was `efytech2` and `#define IO_KEY` was `aio_sXIY05Iri6o3ueuWYL-LY5KnZlCaa`, which were included in the source code `ISS_LED_SAMPLE.ino` before uploading. For location of the ISS, the website <https://www.astroviewer.net/iss/en/> was used.

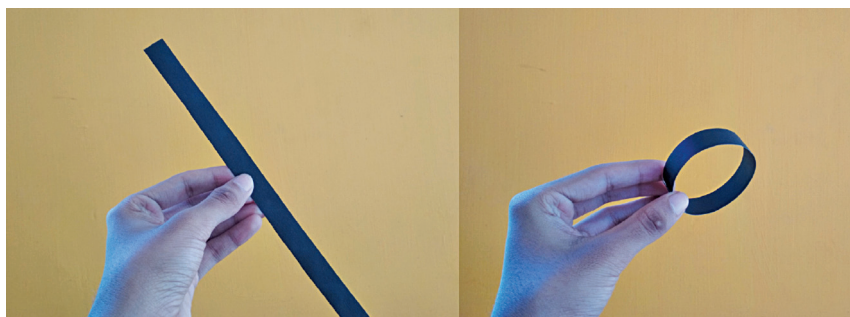


Fig. 13: Making cardboard base of the lamp



Fig. 14: NodeMCU and LED placed at the bottom



Fig. 15: Diffuser placed on top of the base

The circuit

In this last step of getting the project working, connect the LED to D7 pin of your NodeMCU. The 330-ohm resistor in series (as shown in Fig. 12) is used to limit the current through the LED.

To make the base of the lamp, you may cut a strip of black cardboard and roll it into a cylinder. The height of the cylinder (width of the strip) should be a bit more than the height of the NodeMCU when it is resting on its pins. The dia of the cylinder should be the same as that of diffuser's base.

Cut another piece of cardboard in circular shape with its diameter equal to the diameter of the cardboard cylinder and stick the cylinder to it with adhesive. Place the NodeMCU and the LED inside this cardboard base and place the diffuser on top (refer Fig. 13 through Fig. 15).

Your very own ISS overhead indicator is now ready. Every time the ISS passes over your location, the lamp will let you know.

The ISS crosses a location once or twice most days (sometimes at dead of night when it may not get noticed). There are some days when it does not show up at all, but that's rare.

It feels good to know there are people working up there in ISS and it is exciting when they are just overhead. When ISS arrives after sunset, you can even run out and have a look at it moving in the sky and may be wave a hello! **EFY**



Arpan Mondal is pursuing aerospace engineering and is an innovative projects creator. He also likes playing guitar and sketching